FOR REPRESENTABLE MORPHISMS, SURJECTIVITY ON OBJECTS EQUALS SURJECTIVITY

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Source reference. Original reference 03YK: focused-seeds/geometry/S9-crit-representable-by-spaces-surjective/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03ME: focused-seeds/geometry/S9-crit-representable-by-spaces-surjective/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $p : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$. If p is representable by algebraic spaces, then the following are equivalent:*

1. *p is surjective on objects, and*
2. *p is surjective (see Algebraic Stacks, Definition 03YK).*

Proof. Assume (2). Let k be a field and let y be an object of $\mathcal{Y}$ over k. Let X_y denote an algebraic space over k representing the 2-fibre product

$$(Sch/\operatorname{Spec}(k))_{fppf} \times_{y,\mathcal{Y},p} \mathcal{X}.$$

As we've assumed that p is surjective we see that X_y is not empty. Hence we can find a field extension K/k and a K-valued point x of X_y. Via the 2-Yoneda lemma this corresponds to an object x of $\mathcal{X}$ over K together with an isomorphism $p(x) \cong y|_{\operatorname{Spec}(K)}$ and we see that (1) holds.

Assume (1). Choose a scheme T and a 1-morphism $y : (Sch/T)_{fppf} \to \mathcal{Y}$. Let X_y be an algebraic space over T representing the 2-fibre product $(Sch/T)_{fppf} \times_{y,\mathcal{Y},p} \mathcal{X}$. We have to show that $X_y \to T$ is surjective. By Morphisms of Spaces, Definition 03ME we have to show that $|X_y| \to |T|$ is surjective. This means exactly that given a field k over T and a morphism $t : \operatorname{Spec}(k) \to T$ there exists a field extension K/k and a morphism $x : \operatorname{Spec}(K) \to X_y$ such that

$$\begin{array}{ccc} \operatorname{Spec}(K) & \xrightarrow{x} & X_y \\ \downarrow & & \downarrow \\ \operatorname{Spec}(k) & \xrightarrow{t} & T \end{array}$$

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commutes. By the 2-Yoneda lemma this means exactly that we have to find $k \subset K$ and an object x of $\mathcal{X}$ over K such that $p(x) \cong t^*y|_{\mathrm{Spec}(K)}$. Hence (1) guarantees that this is the case and we win. $\square$